

⌵ Semantic Density Analysis

This notebook introduces semantic density as a simple proxy for informational content in political manifestos.

The goal is to explore whether texts differ in how much meaningful content they convey, as opposed to relying on generic or repetitive expressions.

This is particularly relevant in the context of political discourse, where language may appear informative while remaining relatively abstract.

```
import os
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import re
from collections import Counter
```

```
from google.colab import files
uploaded = files.upload()
```

Scegli file

legislatives (1).zip

- legislatives (1).zip(application/x-zip-compressed) - 7384519 bytes, last modified: 28/4/2026 - 100% done

Saving legislatives (1).zip to legislatives (1).zip

Dataset Access

The dataset must be uploaded manually when running this notebook (e.g., in Google Colab), as it is not stored directly within the repository.

This ensures portability and avoids issues related to file paths across different environments.

⌵ 2.1 Loading the dataset

```
import os
import pandas as pd
import zipfile

# Assuming the uploaded zip file is 'legislatives (1) (1).zip'
zip_file_name = list(uploaded.keys())[0]

# Extract the contents of the zip file
with zipfile.ZipFile(zip_file_name, 'r') as zip_ref:
    zip_ref.extractall('/content/')

# Update the folder variable to point to the extracted directory
folder = "/content/text_files/1988/legislatives"

data = []

for filename in os.listdir(folder):
    if filename.endswith(".txt"):
        with open(os.path.join(folder, filename), "r", encoding="utf-8", errors="ignore") as f:
            text = f.read()
            data.append({"id": filename, "text": text})

df = pd.DataFrame(data)

print("Number of documents:", len(df))
df.head()
```

Number of documents: 3628

	id	text	
0	EL174_L_1988_06_003_01_1_PF_03.txt	Sciences Po / fonds CEVIPOF\nELECTIONS LÉGISLA...	
1	EL175_L_1988_06_050_01_2_PF_01.txt	Michel ROCARD 1er Ministre\nvous demande d'app...	
2	EL176_L_1988_06_056_06_1_PF_02.txt	Sciences Po / fonds CEVIPOF\nMajorité présiden...	
3	EL175_L_1988_06_048_01_1_PF_03.txt	Sciences Po / fonds CEVIPOF\nREPUBLIQUE FRANCA...	
4	EL177_L_1988_06_088_03_2_PF_01.txt	Elections législatives du 12 Juin 1988 DÉPARTE...	

Passaggi successivi:

Genera codice con df

New interactive sheet

⌵ 2.2 Text Cleaning

Before computing semantic density, the text is cleaned to reduce noise.

Since the dataset comes from OCR, it may contain irregular characters, broken words, or formatting artifacts. The cleaning process focuses on removing these elements while preserving the main linguistic content.

The goal is not to fully normalize the text, but to ensure that basic word-level statistics are meaningful.

```
def clean_text(text):
    text = text.lower()
    text = re.sub(r"\n+", " ", text)
    text = re.sub(r"^[a-zAÀÇÈÉÊËÏÎÔÛÜŸÑÆ\S]", " ", text)
    text = re.sub(r"\s+", " ", text).strip()
    return text

df["clean_text"] = df["text"].apply(clean_text)
```

2.3 Definition of Semantic Density

Semantic density is defined as the proportion of content-bearing words relative to the total number of words in a document.

The intuition is that texts with higher density contain more specific and informative content, while texts with lower density rely more on general or functional language.

In this implementation, content words are approximated using simple lexical filtering rather than full linguistic annotation.

Although this is a simplification, it remains sufficient for a comparative analysis across documents.

```
def semantic_density(text):
    words = text.split()
    content_words = [w for w in words if len(w) > 3]

    if len(words) == 0:
        return 0

    return len(content_words) / len(words)

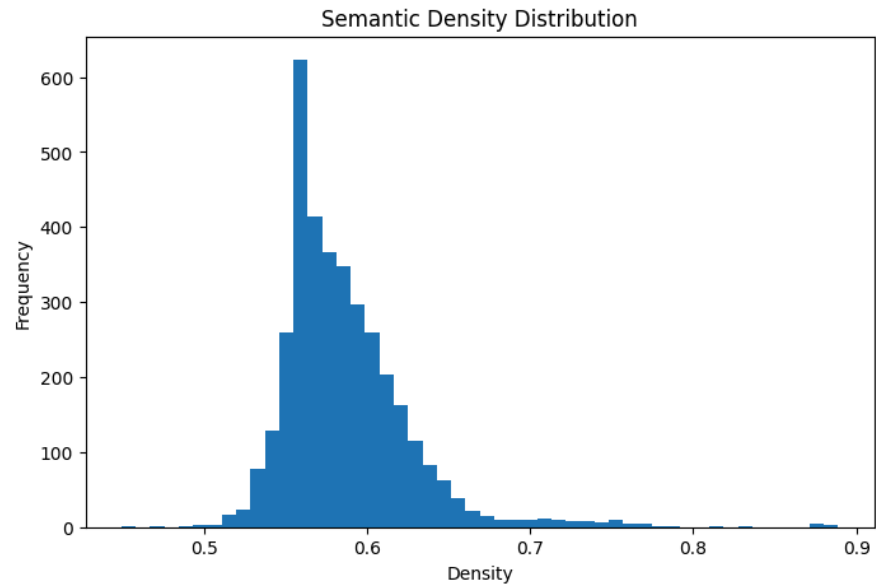
df["density"] = df["clean_text"].apply(semantic_density)
```

2.4 Density distribution

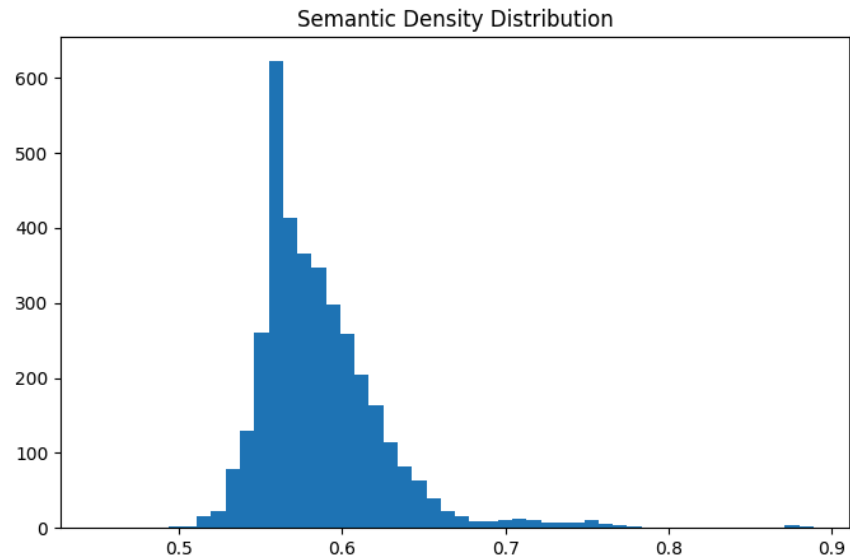
We now compute semantic density for each document in the corpus.

This allows us to compare manifestos in terms of their relative informational content.

```
plt.figure(figsize=(8,5))
plt.hist(df["density"], bins=50)
plt.title("Semantic Density Distribution")
plt.xlabel("Density")
plt.ylabel("Frequency")
plt.show()
```



```
plt.figure(figsize=(8,5))
plt.hist(df["density"], bins=50)
plt.title("Semantic Density Distribution")
plt.savefig("density_plot.png", dpi=300, bbox_inches="tight")
plt.show()
```



Density Distribution

The distribution of semantic density shows that most documents fall within a relatively narrow range, but still exhibit noticeable variation.

Some manifestos appear more content-rich, while others rely more heavily on general or repetitive formulations.

This suggests that political texts differ not only in what they say, but also in how much concrete information they provide.

A closer inspection suggests that lower-density documents tend to use more abstract language, while higher-density documents often include more specific references, such as institutions, local issues, or concrete proposals.

This aligns with the idea of "wooden language", where discourse may appear meaningful but contains limited informational content.

2.5 Density statistics

```
df["density"].describe()
```

	density
count	3628.000000
mean	0.586725
std	0.039760
min	0.449393
25%	0.561003
50%	0.579079
75%	0.603977
max	0.888889

dtype: float64

Summary Statistics

Summary statistics provide a clearer view of the distribution of semantic density across documents.

They help identify typical values as well as potential outliers.

2.6 Quantile analysis

We analyze density distribution using quantiles to better identify extreme documents.

```
df["density"].quantile([0.1, 0.25, 0.5, 0.75, 0.9])
```

	density
0.10	0.550716
0.25	0.561003
0.50	0.579079
0.75	0.603977
0.90	0.630248

dtype: float64

2.7 Low vs High density groups

We define low-density and high-density documents using quantile thresholds.

```
low_threshold = df["density"].quantile(0.1)
high_threshold = df["density"].quantile(0.9)

low_docs = df[df["density"] <= low_threshold]
high_docs = df[df["density"] >= high_threshold]

print("Low density docs:", len(low_docs))
print("High density docs:", len(high_docs))
```

Low density docs: 363
High density docs: 363

```
low_docs[["id", "density"]].head()
```

	id	density	
10	EL174_L_1988_06_013_08_1_PF_02.txt	0.543700	
40	EL177_L_1988_06_094_05_2_PF_02.txt	0.535163	
50	EL176_L_1988_06_071_02_1_PF_02.txt	0.550593	
77	EL177_L_1988_06_092_04_1_PF_03.txt	0.538462	
79	EL177_L_1988_06_076_01_1_PF_01.txt	0.545802	

```
high_docs[["id", "density"]].head()
```

	id	density	
5	EL176_L_1988_06_074_05_2_PF_02.txt	0.639296	
8	EL177_L_1988_06_091_05_1_PF_04.txt	0.647826	
22	EL175_L_1988_06_030_02_1_PF_04.txt	0.639437	
32	EL176_L_1988_06_067_01_2_PF_02.txt	0.640904	
37	EL175_L_1988_06_040_01_2_PF_02.txt	0.650510	

2.8 Qualitative comparison

```
print("LOW DENSITY TEXT:\n")
print(low_docs.iloc[0]["text"][:1000])

print("\n\nHIGH DENSITY TEXT:\n")
print(high_docs.iloc[0]["text"][:1000])
```

LOW DENSITY TEXT:

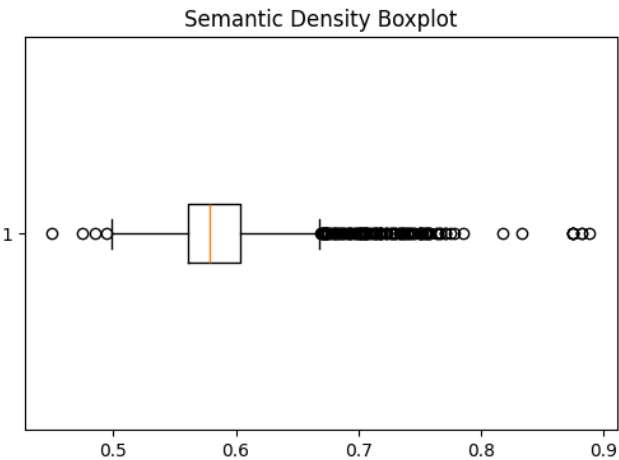
République Française - Election législative du 5 juin 1988
8e CIRCONSCRIPTION
Candidat de Rassemblement des Forces de Gauche présenté par le Parti Communiste Français
Marcel TASSY Adjoint au Maire de Marseille
suppléant : Marcel BENASSI
Madame, Mademoiselle, Monsieur,
Le dimanche 5 juin 88 vous êtes appelés à voter pour élire votre député. Pourquoi si vite ?
Le président de la République l'a expliqué, son gouvernement comprend déjà trois anciens ministres de Giscard d'Estaing, mais il juge c
Le 8 Mai dernier, avec mon parti, pour barrer la route à CHIRAC, BARRE et LE PEN, j'ai contribué à la réélection de François MITTERRAND
Chercher des alliés à Droite, ce ne peut pas être pour faire une politique de Gauche. On n'a jamais vu la Droite soutenir aut

HIGH DENSITY TEXT:

Sciences Po / fonds CEVIPOF
MAJORITÉ PRÉSIDENTIELLE POUR LA FRANCE UNIE
Mobilisons-nous le 12 Juin contre l'alliance Mazeaud / Front National
Donnons à FRANÇOIS MITTERRAND les moyens de sa politique
VOTEZ pour que le Chablais et le Giffre donnent un député à MICHEL ROCARD
SERGE DUPESSEY
Maire Adjoint d'Evian Marié, 4 enfants, chirurgien-dentiste né à Thonon
FRANÇOISE CHAMBLAS Mariée, 2 enfants née et domiciliée à Thonon enfance à Larringes Professeur au C.E.S. de Publier
UNIR LA FRANCE UNIR LE CHABLAIS ET LA VALLÉE DU GIFFREMAJORITÉ PRÉSIDENTIELLE POUR LA FRANCE UNIE
Habitantes, habitants du Chablais et de la Vallée du Giffre, mobilisons nous ...
... "Progressons sur le chemin de l'égalité ". F. MITTERRAND - MAI 1988
Le 12 Juin
VOTEZ ET FAITES VOTER
• Contre la course aux armements
• Contre la main-mise de l'Etat par un groupe partisan
• Contre le racisme, la xénophobie et toutes les formes d'exclusion
• Contre l'installation, dans notre région, d'une droite dure Mazeaud - Front Nation

2.9 Boxplot visualization

```
plt.figure(figsize=(6,4))
plt.boxplot(df["density"], vert=False)
plt.title("Semantic Density Boxplot")
plt.show()
```



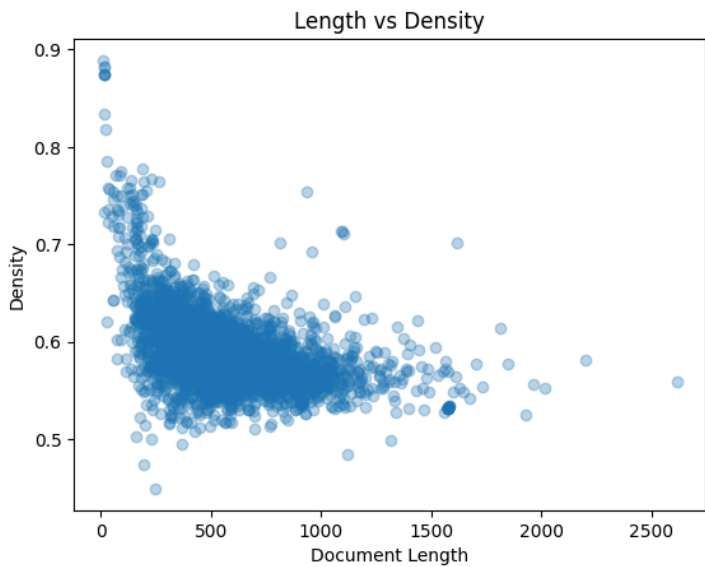
2.10 Density vs document length

```
df["length"] = df["clean_text"].apply(lambda x: len(x.split()))

df[["density", "length"]].corr()
```

	density	length	
density	1.000000	-0.420071	
length	-0.420071	1.000000	

```
plt.scatter(df["length"], df["density"], alpha=0.3)
plt.xlabel("Document Length")
plt.ylabel("Density")
plt.title("Length vs Density")
plt.show()
```



2.11 Lexical analysis

```
low_words = " ".join(low_docs["clean_text"]).split()
high_words = " ".join(high_docs["clean_text"]).split()

print("Low density words:", Counter(low_words).most_common(10))
print("High density words:", Counter(high_words).most_common(10))
```

Low density words: [('de', 13995), ('la', 8666), ('et', 7368), ('l', 6188), ('le', 6119), ('à', 5356), ('les', 4881), ('des', 4439), ('
High density words: [('de', 5299), ('la', 3108), ('et', 2653), ('le', 2077), ('l', 1777), ('pour', 1769), ('du', 1679), ('à', 1480), ('

2.12 Interpretation

Semantic density provides a simple but useful way to compare documents in terms of informational content.

Although the measure is approximate, it highlights meaningful differences across political manifestos.

These results motivate further analysis using semantic similarity methods, which can capture relationships between texts beyond word-level statistics.

Although simple, this measure already captures a dimension of political discourse that is not immediately visible through basic frequency analysis.

It provides a bridge between purely lexical approaches and more advanced semantic methods.